

Lecture 4 ERS312 Notes

2nd Ocean Exploration

- Snails found in caves used as food
- Reasons for ocean use as a food resource (climate) *ups and downs of temperature which is known as ice ages
- People looking for alternate food sources

Pacific navigators

- People were settling in tropical islands (very small)
- Travel taken place in small boats in a long distance
- Only a few made it to islands
- Archeological evidence shows people came from New Guinea

Thor Heyerdahl

- To test theory if occupation in the Pacific Island could have occurred from South America
- This raft set out from Peru and travelled with the ocean currents (took 13 weeks)
- Even though test was successful DNA shows that Islanders come from west

Early European Navigators

- Phoenicians came from western Asia
- Trades people (to trade goods they developed navigation)
- Also travelled to Indian ocean

World Map of Herodotus

- Produced the first world map
- He didn't portray Africa very well
- Helped people navigate oceans

Eratosthenes

- Did experiment: looked at sun and looked at shadow sun cast at 2 different spots on earth
- Was a well at syene where the sun rays hit the bottom of the well once a year (was directly above well *sun was in the zenith June 22 (summer solstice) *Tropic of Cancer
- If there's an angle of 7 degrees (distance between Alexandria and syene represents 1/50 of the circumference of the earth)
- They walked and measured length between two distances

Estimation of circumference of Earth *maybe question on test

- $1/50$ of a circle = 5000 stadia
- So 1 circle is 50×5000 stadia = 250 000 stadia (approx.. 40 000km)

Ptolomy produced accurate world map

- Was most advanced world map at the time
- Knowledge of western world had increased a bit
- Also wrote a book called Geography (he reported earths circumference as 30 000km instead of Eratosthenes who said it was 40 000km)

Middle Ages-Arabian Navigators

- Used Indian monsoon winds (reversed in direction) to navigate and trade
- The Vikings explored north atlantic oceans during this time (was rough weather and sea ice)
- Did exploration on land aswell (best known for ocean voyages)
- Travelled to Iceland then Greenland

Viking Routes and Colonies

- Settled in Iceland
- From there they kept on exploring, southern coastlines of Greenland
- Greenland tricked Vikings to settle in Greenland (southern greenland was somewhat green)
- Vikings made it to North America

Northern Hemisphere Temperatures since 700

- Temperatures are going up steeply
- Warming trend (900-1200) and progressive cooling trend (1600-1800)
- Vikings abandoned Greenland by 1500 because it was getting too warm

Voyages of Columbus and Magellan

- People wanted to find ways to trade especially in Asia
- Sultan Mohammed II captured Constantinople which is now Istanbul
- Magellan was killed near Philippines

Toscanelli World map with wrong circumference

- Columbus believed circumference of Earth to be 30000km
- He thought he came to an island in Asia (because of using the wrong map) *he was in North America

John Cabot

- Travelled across England (set foot on Canada)
- Came south of Newfoundland (he discovered abundant cod at Grand Banks)
- Salted cod (how the fish was preserved)

James Cook's Voyages

- Did a lot of mapping
- Spent years at sea, 3 around the world voyages
- Produced accurate maps of coastal areas (had access to technology) *chronometer was developed in 1771
- Amongst the first to determine longitude
- Traded goods he acquired in Hawaii

Darwin's voyage on HMS Beagle

- Found that each island had different beaks of finches

The Northwest Passage

- Lot of interest finding a trade mark
- Why would you consider travel from north despite it being frozen
- Exploration happened during summer
- Spending the winters in Canada

The lost Franklin Expedition

- Stronger ships
- The expedition perished (British decided to find them) *led to a lot of discovery in Canada
- Invested a lot to find evidence of franklin expedition
- In 2014 found ship

Discovery of the NW Passage in 1906

- Amundson was famous Norwegian explorer
- He wintered at an Inuit settlement (learned how to survive the ice)

Modern day oceanography

Titanic shipwreck

- Was thought as being unsinkable ship
- The route is a very common ocean route for goods
- Led people to think to make observations of Icebergs
- Can see iceberg alley from Greenland (they break off and come off via glaciers)
- Icebergs were bigger because Ice was colder in 1912
- It was the advent of International Ice Patrol

Bridge of modern research vessel

- Steering done by joysticks
- Drifter Buoy (placed in water and drifted)
- Niskin Rosetta for Water Sampling (device that has individual tanks on it and they're open both on top and bottom; placed in water and slowly goes down to deep sea) *each bottle

closes at a certain ocean depth, so you collect water for each specific point of the ocean you are at

- Seafloor Lander for Longterm observations
- Plankton Tow Net (to study plankton)
- Side Scanning Sonar (could calculate depth of the ocean basins) *image the sea floor